PRODUCT USER MANUAL

For North Pacific Gyre Oscillation OMI_VAR_EXTREME_CLIMVAR_PACIFIC_npgo_sla _eof_mode_projection

Issue: 4.0

Contributors : C Boone

Approval Date : November 2025

MERCATOR OCEAN
INTERNATIONAL
2 avenue de l'Aérodrome de Montaudran, 31400 Toulouse, FRANCE
Tél : +33 5 61 39 38 02 - Fax : +33 5 61 39 38 99
Société civile de droit français au capital de 2 000 000 € - 522 911 577 RCS Toulouse - SIRET 522 911 577 00024

marine.copernicus.eu
mercator-ocean.eu

RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	07/10/2019	all	Creation of the document	J.-F. Legeais	C. Derval
1.1	01/04/2020		Typo correction	J.-F. Legeais	C. Derval
1.2	20/08/2020	I1, II1, II2	CF convention	J.-F. Legeais	C. Derval
2.0	2022/04/04	all	Use new template ; temporal extension;	C Pegliasco	C. Derval
2.1	05/04/2023	I.2	Extension of the time series	MI Pujol	C. Derval
3.0	15/06/2023		OMI nomenclature change Use « myint » dataset	MI Pujol	
4.0	28/05/2025		Update version nov2025 : Temporal extension DT24 ; Use new template	C Boone	Copernicus Marine Product Management

TABLE OF CONTENTS

RECORD TABLE **2**

GLOSSARY AND ABBREVIATIONS **4**

DATA ACCESS **5**

1) INTRODUCTION **6**

a) Summary 6

2) DESCRIPTION OF THE PRODUCT SPECIFICATION **7**

a) General Information 7

b) Production System Description 8

c) Dataset details 8

3) FILE FORMAT **9**

4) FILES NOMENCLATURE **9**

5) REFERENCES **10**

GLOSSARY AND ABBREVIATIONS

MOI	Mercator Ocean International
NetCDF	Network Common Data Form
OMI	Ocean Monitoring Indicator
PUM	Product User Manual
QUID	Quality Information Document

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

This document is the user manual for the CMEMS Ocean Monitoring Indicator (OMI) OMI_VAR_EXTREME_CLIMVAR_PACIFIC_npgo_sla_eof_mode_projection. It contains 1 OMIs :

- OMI_VAR_EXTREME_CLIMVAR_PACIFIC_npgo_sla_eof_mode_projection

The OMI provides a time series of the sea level North Pacific Gyre Oscillation (NPGO) index from 1950 until the present.

The product is published on the CMEMS dissemination server after automatic and delayed-mode quality controls. The product is available online and disseminated through the CMEMS Information System. Files downloaded are in NetCDF format and follow CF-1.7 convention.

Disclaimer: The quality of the product may vary during the proposed time series depending on the possible update of the system.
The North Pacific sea level NPGO product is described in the Quality Information Document (QUID) given in section **Erreur ! Source du renvoi introuvable.**. See also News flash.

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

Product Lines	OMI_VAR_EXTREME_CLIMVAR_PACIFIC_npgo_sla_eof_mode_projection
Geographical coverage	PACIFIC
Variables	time npgo : The North Pacific Gyre Oscillation (NPGO) is a climate pattern Di Lorenzo et al., « North Pacific Gyre Oscillation Links Ocean Climate and Ecosystem Change ». defined as the second dominant mode of variability of Sea Surface Height (SSH) anomaly in Northeast Pacific (25°–62°N, 180°–250°E). From the principal component analysis, the EOF's amplitude pattern or correlation map has been inferred and projected on the SSH anomalies from an historical eddy-permitting model hindcast of the Northeast Pacific over the period 1950-2004.
Available time series	[1950, - 6M to 1M]
Temporal resolution	Monthly
Horizontal resolution	
Integration depth (for example)	Sea surface
Format	NetCDF4

b) Production System Description

From 1950 to 2003, the NPGO index is derived from numerical model analysis (Di Lorenzo et al, 2008).
From 2004, the CMEMS NPGO OMI is derived from:

- The DUACS (https://duacs.cls.fr) delayed-time (reprocessed version DT-2024, see https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-SL-QUID-008-032-068.pdf) altimeter sea level gridded product: CMEMS SEALEVEL_GLO_PHY_L4_MY_008_047
- The DUACS near real-time altimeter sea level gridded product: CMEMS SEALEVEL_GLO_PHY_L4_NRT_008_046.

c) Dataset details

Omi_var_extreme_climvar_pacific_npgo_sla_eof_mode_projection contains the monthly evolution of the sea level in the North Pacific Gyre Oscillation derived since 2004 from the altimeter gridded sea level maps where the 2nd EOF have been projected. The annual, semi-annual periodic signals and the trend are adjusted. The NPGO index is centred and normalised to fit with the Di Lorenzo index estimated over the [1993, 2003] period (Di Lorenzo et al., 2008). Finally, the NPGO index deduced from altimetry is completed with historical index from Di Lorenzo et al (2008) from 2004 downward.
time [days since 1950-01-01 00:00:00] Time, gregorain calendar **npgo [1]** longname : ."Dimensionless North Pacific Gyre Oscillation indicator" standard_name : ."tendency_of_sea_surface_height_above_mean_sea_level"

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

OMI files nomenclature when downloaded through the CMEMS FTP Services is based on the following nomenclature:

<datasetname>_<timestep>_P<productiontime>.nc

Where :

- **<datasetname>** is the dataset reference .
- **<timestep>** is the first time step contained in the file.
- **<productiontime>** is the production time when the file was generated.
- **.nc** is the standard NetCDF filename extension.

Example:

OMI_VAR_EXTREME_CLIMVAR_PACIFIC_npgo_sla_eof_mode_projection_9500115_P20220331.nc

5) REFERENCES

Quality Information Document (QUID)
http://marine.copernicus.eu/documents/QUID/CMEMS-OMI-QUID-VAR_EXTREME-CLIMVAR-NPGO.pdf

Di Lorenzo, E., N. Schneider, K. M. Cobb, P. J. S. Franks, K. Chhak, A. J. Miller, J. C. McWilliams, et al. « North Pacific Gyre Oscillation Links Ocean Climate and Ecosystem Change ». *Geophysical Research Letters* 35, n° 8 (2008). https://doi.org/10.1029/2007GL032838.